What should AI do when key campaign details are missing?

Social Campaign Brief



Product / Focus
Reusable bottle

Target audience
Local students



Budget	Not provided
Channel	Not provided
Success measure	Not provided

What problem are we solving?

Learn how to test and fix a prompt: run it with complete and incomplete inputs, find where it invents information, and add one fallback instruction until it handles missing facts safely.

THE SITUATION

The problem

Joon is a marketing intern at a small shop that sells eco-friendly products. His manager asked for a two-week social media plan for a new reusable bottle. The brief did not include the budget, the channel, or the success measure. Joon's root problem is that he must deliver a plan quickly, but key facts are missing. His idea is to let ChatGPT write the plan. The plan looked ready to approve: "Budget: $5,000. Channel: influencer ads. Goal: 1,000 sales." All three numbers and choices were invented. The real budget, which the manager gave later, was $300. The AI filled the gaps because the prompt never told it what to do when information is missing.

Today's goal and a three-step method

Learn how to test and fix a prompt: run it with complete and incomplete inputs, find where it invents information, and add one fallback instruction until it handles missing facts safely.

STEP 1

Test with different inputs

Run the prompt with a complete brief and with briefs that miss information.

STEP 2

Find the bug

Count the details in the answer that are not in the input. Each invented detail is a bug.

STEP 3

Patch and retest

Add one fallback instruction for missing information, then run all the tests again.

Campaign brief

Fictional data created for this class. File: data/session_4_2_campaign_brief.csv

Field	Value
Product	Reusable water bottle, $18 each
Audience	Local university students
Timing	Two weeks
Budget	Not provided
Channel	Not provided
Success measure	Not provided

Test inputs for checking the prompt

Fictional data created for this class. File: data/session_4_2_campaign_brief.csv

Test	Budget	Channel	Success measure
Test 1	Not provided	Not provided	Not provided
Test 2	$300	Not provided	100 visits to the product page
Test 3	$300	The shop's Instagram account	Not provided

Five testing words you need first

Before you fix a prompt, you must know these words. Each example uses Joon's campaign brief.

KEY IDEA

Test input

One version of the brief that you use to try the prompt. Example: Test 2 has a budget but no channel.

KEY IDEA

Bug

A repeated wrong behavior in the answer. Example: the AI invents a budget whenever the budget is missing.

KEY IDEA

Fallback instruction

A rule for what the AI must do when information is missing. Example: "If the budget is missing, write 'Missing: budget' and ask for it."

Five testing words you need first (continued)

Before you fix a prompt, you must know these words. Each example uses Joon's campaign brief.

KEY IDEA

Patch

A small change to the prompt that fixes one bug. Change one instruction at a time, so you know what helped.

KEY IDEA

Retest

Run all the test inputs again after a patch. A fix for one input must not break another.

Why the AI plan was a problem

"Looks ready to approve" and "uses only the facts" are different questions. Compare the AI plan with the brief.

AI plan said	The brief says	Problem
Budget: $5,000	Not provided	Invented number
Channel: influencer ads	Not provided	Invented choice
Goal: 1,000 sales	Not provided	Invented target

KEY POINTS

What the table shows

- 3 of 3 missing fields were filled with invented details, and none was marked as a guess.
- The invented budget was more than 16 times the real budget: $5,000 / $300 = 16.7.
- The AI was not careless. Without a fallback instruction, it treats "missing" as "make a reasonable guess."

How the three-step method fixes the prompt

The instructor runs Poor Prompt 1 and Improved Prompts 1 and 2 below in front of the class and compares the answers.

KEY IDEA

Step 1 fixes the hidden-bug problem

A prompt that works on one complete brief may fail on an incomplete one. Testing several inputs shows the bug.

KEY IDEA

Step 2 fixes the "looks fine" problem

Counting invented details turns a feeling into a number you can compare after the patch.

KEY IDEA

Step 3 fixes the bug safely

One fallback instruction, then a retest of all inputs, shows whether the patch worked without breaking anything.

Poor prompt 1

Same case. Compare what the prompt asks for.

POOR PROMPT 1 - DO NOT USE

AI Chat

You

Plan a two-week social media campaign for our new reusable bottle. We want more local students to learn about it.

Message AI...

WHAT THE AI OFTEN ANSWERS

A typical result

A complete-looking plan with a budget of several thousand dollars, influencer ads, daily posts, and a sales target.

WHY IT IS WEAK

Reasons

- It does not list which facts are known and which are missing.
- It has no fallback instruction, so the AI fills every gap with a guess.
- The guesses look like facts, so a manager could approve the wrong budget.

Poor prompt 2

Same case. Compare what the prompt asks for.

POOR PROMPT 2 - DO NOT USE

AI Chat

You

Plan a two-week social media campaign for our new reusable bottle for local
university students. Make it complete and ready to approve.
Fill in any missing details with your best numbers.

Message AI...

WHAT THE AI OFTEN ANSWERS

A typical result

A plan with an exact budget, channel, and sales goal, each presented as decided.

WHY IT IS WEAK

Reasons

- "Fill in missing details with your best numbers" tells the AI to invent facts.
- "Ready to approve" hides the fact that the team has not decided the budget or channel.
- Nothing in the answer shows which parts are real and which are guesses.

Improved prompt 1

Steps 1 and 2. Copy this prompt exactly.

IMPROVED PROMPT 1 - COPY AND SEND

AI Chat

You

You help plan small social media campaigns. Use only the brief below.

Brief:
- Product: reusable water bottle, $18 each
- Audience: local university students
- Timing: two weeks
- Budget: not provided
- Channel: not provided
- Success measure: not provided

Before you plan anything, check each field and mark it Provided or Missing.
For each missing field, write "Missing: [field]" and ask one question about it.
Do not invent any budget, channel, goal, or number. Do not write the plan yet.

Message AI...

Why it works: improved prompt 1

Each part of the prompt matters.

WHY IT WORKS

Reasons

- It lists every field, so the AI can see exactly what is missing.
- "Mark it Provided or Missing" makes the gaps visible before any planning.
- "Missing: [field]" is the fallback instruction that fixes the bug in Poor Prompts 1 and 2.
- One question per missing field gives Joon a short list to ask his manager.
- "Do not write the plan yet" keeps checking and planning as separate steps.

Improved prompt 2

Same case. Compare what the prompt asks for.

IMPROVED PROMPT 2 - COPY AND SEND

AI Chat

You

My manager answered two questions: the budget is $300, and the success
measure is 100 visits to the product page in two weeks.
The channel is still not decided.

Now draft a two-week plan that stays within $300.
Show how the $300 is split; the total must be $300 or less.
Suggest two channel options and label each one "Option to approve."
Label any number you estimate as "Estimate."

Message AI...

WHY IT WORKS

Reasons

- It adds only the facts the manager gave, so the plan uses real numbers.
- "The total must be $300 or less" gives a limit you can check with a calculator.
- "Option to approve" keeps the undecided channel as a choice, not a fact.
- The "Estimate" label shows which numbers are guesses.

Exercise 1: Find the bug

Guided practice. Follow the steps in order.

GUIDED PRACTICE

Steps

1. Open ChatGPT and start a **new chat**. Copy Poor Prompt 1 from the box above, paste it, and send it. Count the budget, channel, and goal details in the plan that are not in the brief. You will need the count for your report.
2. Start another **new chat**, so the first answer does not affect the second. Copy Improved Prompt 1, paste it, and send it.
3. Compare the AI answer with the answer key below. Check that no budget, channel, or goal was invented.
4. In the same chat, copy Improved Prompt 2, paste it, and send it. Add up the budget split with a calculator and check that the total is $300 or less, and that the channels are labeled "Option to approve."
5. For the Results section of your report, write the bug count for Poor Prompt 1 (for example, "3 invented details") and the count for Improved Prompt 1 (it should be 0).

Answer key

Use this table to check the AI.

Field	Status	A good question to ask
Product	Provided	None
Audience	Provided	None
Timing	Provided	None
Budget	Missing	How much can we spend in two weeks?
Channel	Missing	Which social media account should we use?
Success measure	Missing	What result will show that the campaign worked?

Exercise 2: Patch and retest

Applied practice with a template.

HOW IT WORKS

Read first

- A good prompt must work on every test input. Use this template to run Test 2 and Test 3. Replace each part in square brackets.

TEMPLATE - FILL IN, THEN SEND

AI Chat

You

You help plan small social media campaigns. Use only the brief below.
Brief: Product: reusable water bottle, $18 each. Audience: local university students.
Timing: two weeks. Budget: [budget]. Channel: [channel]. Success measure: [measure].
If a field is not provided, write "Missing: [field]," ask one question about it,
and do not invent it.
Draft a two-week plan using only the provided fields.
Show how the budget is split. Label any estimate "Estimate."

Message AI...

Part A: Worked example (run it)

Copy and send this completed prompt in the same chat.

PART A - COPY AND SEND

AI Chat

You

You help plan small social media campaigns. Use only the brief below.
Brief: Product: reusable water bottle, $18 each. Audience: local university students.
Timing: two weeks. Budget: $300. Channel: not provided.
Success measure: 100 visits to the product page.
If a field is not provided, write "Missing: [field]," ask one question about it,
and do not invent it.
Draft a two-week plan using only the provided fields.
Show how the budget is split. Label any estimate "Estimate."

Message AI...

CHECK THE ANSWER

Expected result

Case: Start a **new chat**, then copy and send this prompt for Test 2.

The plan writes "Missing: channel" and asks one question about it. If it suggests channels, they are options, not decisions. The budget split adds up to $300 or less, and the goal stays 100 product-page visits. The bug count is 0.

Part B: Your turn (write it)

Fill in the template yourself and send it.

YOUR TASK

Write your own prompt

Run Test 3: the budget is $300, the channel is the shop's Instagram account, and the success measure is not provided. Fill in the template yourself and send it in a new chat.

CHECK THE ANSWER

Expected result

The plan uses Instagram and stays within $300. It writes "Missing: success measure" and asks one question, such as "What result will show that the campaign worked?" It must not invent a goal such as "1,000 sales." The bug count is 0.

Write your session report

Use Report_template.docx in Microsoft Word. Save it as FirstName_StudentID_W14_Keyword.docx and upload it to iLearn by 23:59 on class day.

SUBMIT

One report for this session

- **Introduction:** Explain Joon's root problem (a plan is needed, but key facts are missing), his idea to let AI write it, and what went wrong with the first plan.
- **Method:** Copy Poor Prompt 1, Improved Prompts 1 and 2, the Part A prompt, and the Part B prompt you wrote.
- **Results:** Include a test log with one row per test: the input, the bug count, and what the AI did with the missing field. Add your budget check for Improved Prompt 2.
- **Conclusion:** Recommend the fallback instruction Joon should keep in his prompt, in one or two sentences. Name one result that supports it and one risk that remains.

How to know the work is complete

Check your report before you hand it in.

VERIFY

Success check

- You can explain in one sentence why the $5,000 plan was dangerous, and your Part B prompt produces a plan with "Missing: success measure" and a bug count of 0.